Analyzing Systems Using Data Dictionaries

CSE 4407

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The Data Dictionary

- What
  - Think of it as a specialized dictionary, but for *data*
  - A reference work of **metadata**
  - Example: Customer ID, Order Status Code, Product Price, etc.
  - “The system’s dictionary of data terms.”
- Core Purpose
  - Guides analysts during analysis and design
  - Collects, coordinates, and confirms data term meanings across the organization
  - Ensures everyone speaks the same “data language”

The Need for a Data Dictionary

- Automated Tools?
  - Many DBMS and CASE tools include automated DDs
  - They catalog items, provide templates, ensure uniformity
- Crucial Role: Keeping data CLEAN!
  - Ensures data consistency $\rightarrow$ Avoids chaos!
  - Example: External Customer stored as “Ext”, “E”, or “1”? $\rightarrow$ Bad!
  - “A DD is like a data ‘style guide’ - keeps things tidy!”
- Vital for Analyst
  - Even with tools, *you* need to know:
    - What goes into a DD?
    - Conventions and structure
    - How it’s developed $\rightarrow$ Often starting from DFDs!
  - Helps conceptualize the system deeply

Key Uses of a Data Dictionary

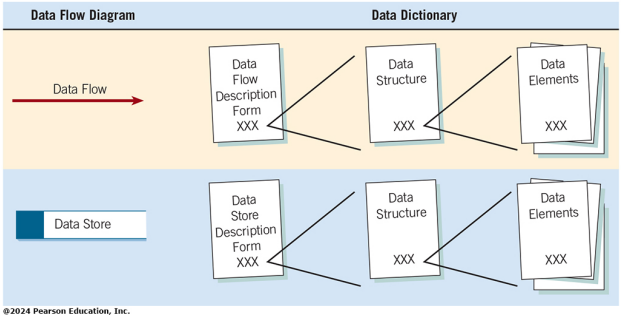
- Validate DFDs: Ensure completeness and accuracy $\rightarrow$ Does every flow/store/process have defined data?
- Develop Screens and Reports: Provides a starting point for fields needed
- Define Data Stores: Determines the exact contents of files/tables
- Develop Process Logic: Helps create detailed logic (mini-specs) for DFD processes
- Create XML: Structures data definitions for data exchange

Data Repository vs. Data Dictionary

- Data Dictionary: Focuses on data definitions and procedures
- Data Repository: A *larger* collection of project information $\rightarrow$ Project Encyclopedia
- Inside a Repository
  - Procedural Logic and Use Cases
  - Screen and Report Designs
  - Data Relationships
  - Project Requirements and Deliverables
  - Project Management Info
- A Data Dictionary focuses only on data, while a Data Repository serves as a central hub for all project-related information.

How DD Entries are Born

- Creating the Dictionary: Examine DFDs!
- Steps
  - Identify a Data Flow or Data Store on the DFD
  - Create a basic Description Form for it
  - Define its Data Structure
  - Define each individual Data Element



Demo: Customer Order Confirmation Example

Data Flow Definition: The System's Conversations

- Data Flows: Represent data moving in the system (inputs, outputs, internal)
- Starting Point: Define system inputs/outputs first
- How: Interviews, Observation, Analyzing Documents/Existing Systems
- Key Info to Capture (per flow)
  - Unique Name
  - Source and Destination
  - Type
  - Associated Data Structure(s)
  - Volume and Comments

Data Flow Description

ID

Name: Customer Order

Description: Contains customer order information and is used to update the customer master and item files and to produce an order record.

Source	Destination
Customer	Process 1

Type of Data Flow

☐ File ☑ Screen ☐ Report ☐ Form ☐ Internal

Data Structure Traveling with the Flow	Volume/Time
Order Information	10/hour

Comments: Order record information for one customer order. The order may be received by Web entry, email, FAX, or by the customer telephoning the order-processing department directly.

Data Structures: The Blueprint

- Data Structure: Defines *what* elements make up a data flow or data store
- Method: Algebraic Notation
  - = means “is composed of”
  - + means “and”
  - { } means repetition
  - [ ] means either/or
  - ( ) means optional
- Structured Records: Group related elements
- Goal: Break down structures into their basic Data Elements

Demo: Data Structure Description Example

Data Structures: Logical vs. Physical

- Two Views
  - Logical
    - The *user's view* or business perspective
    - What data does the business need? $\rightarrow$ Customer Name, Item Price
    - Reflects user's mental model
  - Physical
    - The *implementation view*
    - Includes elements needed for the system to work efficiently $\rightarrow$ Keys, Status Codes, Passwords

Customer Billing Statement	=	Current Date + Customer Number + Customer Name + $\overset{5}{1}${Order Line} + (Previous Payment Amount) + Total Amount Owed + (Comment)
Order Line	=	Order Number + Order Date + Order Total

Data Elements: Foundation

- What: The smallest meaningful unit of data $\rightarrow$ Cannot be broken down further
- Define ONCE: Each element gets a single, authoritative definition in the DD
- Key Characteristics
  - Element Name (Unique, descriptive)
  - Aliases (Synonyms used elsewhere)
  - Description (Clear meaning)
  - Length and Data Type
  - Validation Rules (How to ensure it is correct)
  - Base or Derived?

Demo: Element Description Form Example

Data Elements: Core Attributes

- Length: How much space does it need?
  - Numeric
    - Based on largest possible value
    - Room for growth
  - Text
    - Use standards/sampling (Book)
    - Trade-offs if too short
- Type: What kind of data is it?
  - Numeric, Character/Varchar, Date, Currency, Binary, etc.
  - Choice depends on usage (display, calculation)
- Format: How is it displayed/entered?
  - Uses symbols: `9` for number, `X` for any character, `Z` for zero suppress, and `,./-` for editing
  - Example: `999-99-9999` for Social Security Number in US

Data Elements: Advanced Characteristics

- Validation Criteria: Ensuring data accuracy
  - Discrete
    - Fixed values (e.g., Color Codes: ‘BL’, ‘WH’, ‘GR’)
    - Use List or Table
  - Continues
    - Range of values (e.g., GPA: 0.00 - 4.00)
    - Use Range or Limits
  - Check Digits (for key fields)
- Base vs. Derived
  - Base
    - Entered directly (e.g., Customer Name, Price)
    - Must be stored
  - Derived
    - Calculated by a process (e.g., Order Total, Age)
    - May or may not be stored
- Default Value: Pre-filled value to speed up entry (e.g., Default State)
- Aliases: Listing synonyms (e.g., `Customer Number` = `Client Number`)

Data Stores: Where Data Lives

- What
  - Where the system's data resides persistently
  - Files, databases, tables
  - Group related elements into structural records (often based on entities like `Customer`, `Product`, `Order`)
  - May need to combine info from *multiple* data flows
- Purpose: Stores collections of base elements (and sometimes derived)
- Key Info
  - ID and Name
  - File Type and Format
  - Size/Growth
  - Primary and Secondary Keys

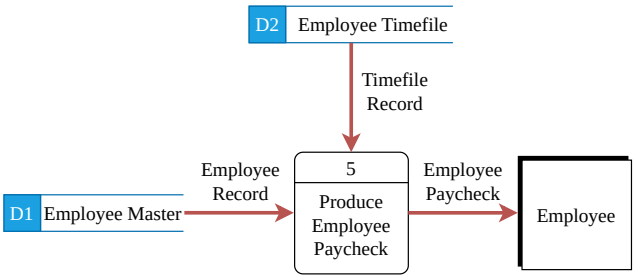
Demo: Data Store Description Form Example

Creating and Data Dictionary: When and How

- Flexible Timing
  - Option 1: Create *after* DFDs are finalized
  - Option 2: Create *concurrently* with DFDs (Top-Down)
- Top-Down Approach (Recommended)
  - Initial Interviews $\rightarrow$ Sketch Diagram 0
  - Create initial DD entries $\rightarrow$ High-level Flows and Structures
  - More Analysis (Interviews, Observations, Docs) $\rightarrow$ Refine DFDs (Child Diagrams)
  - Expand DD with detailed Structures and Elements
- Analogy: Like designing a building!

DFD Levels and Data Dictionary Consistency

- Crucial: DD detail MUST match the DFD level it describes
- Balancing Act
  - Any data structure or element flowing into or out of a child process...
  - ...MUST be contained within a data flow shown on the parent diagram for that process



Employee Record = Employee Number + Personal Information + Wage Information + Current Pay Information + Year-to-Date Information

Timefile Record = Employee Number + Employee Name + Hours Worked

Employee Paycheck = Employee Number + Employee Name + Address + Current Pay Amounts + Year-to-Date Figures

Step 1: Analyzing Inputs and Outputs

- Systematically identify and categorize ALL system inputs and outputs
- Why: Often define user interactions and core data requirements
- Tool: Input/Output Analysis Form
- Key Info
  - Descriptive Name
  - User Contact
  - Input or Output?
  - Format
  - Sequencing Elements
  - Element List

Demo: Input and Output Analysis Form Example

Step 2: Identifying Structures from I/O Analysis

- Mine the I/O Forms for Structures!
- Examine the element list closely
- Look for
  - Elements that logically belong together → First Name, Last Name, Middle Initial
  - Repeating elements or groups → Multiple order lines
  - Optional elements → Apartment Number
  - Mutually exclusive elements → Different payment type details
- Action: Group related elements into **Structural Records** in DD

Step 3: Developing Data Stores

- Focus: Data at rest
- Purpose: Hold permanent or semi-permanent information needed beyond a single transaction
- How to choose
  - Analyze processes
  - What info is temporary?
  - What needs to be kept long-term?
- What to store
  - Primarily store base elements
  - Derived data?
    - Calculated values *don't have to be* stored; can often be recalculated
    - Storing them is a design choice → Storage space vs. calculation time

Demo: Data Store Example

Data Stores vs. User Views

- Data Stores
  - Aim to store data efficiently, often normalized, representing core entities
  - Data at rest
- User Views
  - Structures designed specifically for how a user wants to see data on a particular report on screen
  - Data presentation
- Relationship: Might pull data from multiple data stores and present it in a specific, combined format
- Analogy
  - Data Stores = A library with books organized by genre, author, and title.
  - User View = A custom reading list you've created for a weekend getaway, which includes books from various genres that fit the mood you want.

Data Dictionary: The Payoff!

- Make DD Work for You
  - Ideal DD: Automated, Interactive, Online, Evolutionary
  - NOT just a document, a living tool
- Golden Rule: KEEP IT CURRENT!
  - Must be updated whenever the system design changes
  - Outdated DD is worse than useless $\rightarrow$ Misleading
- Perspective: View DD creation as an activity that parallels analysis and design, not a separate, final task

Use Case 1: Creating Screens, Reports, and Forms

- Definition to Design: DD is a direct input for UI/Report design
- How
  - DD provides the elements needed and their lengths
  - Arrange these elements logically on the screen/page using design principles
- Mapping Structures to Layout
  - Structural Records: Group related fields together visually
  - Repeating Groups: Often become columns in a table or list on the layout

Demo: Order Picking Slip Data Structure vs Order Picking Slip

Use Cases 2 and 3: Generating Code and Analyzing Design

- **User Case 2: Generating Code Structures**
  - DD Definitions can directly generate:
    - Database table schemas
    - Program code structures
  - Ensures code matches the defined data structures
- **Use Case 3: Analyzing System Design:** Use DD to find logical flaws in your DFDs
  - Base Elements Rule: Base elements appearing in an output flow *must* have been input to that process
  - Derived Elements Rule: Derived elements *must* be created by a process
  - Data Store Rule: Elements flowing into/out of a data store *must* exist within that data store's definition in the DD

Use Case 4: Creating XML

- What: e**X**tensible **M**arkup **L**anguage
  - The Universal Data Translator
  - Primary Use: Exchanging data between businesses, systems, or software
- Key Concept: Content vs. Format
  - HTML $\rightarrow$ Defines how data looks like
  - XML $\rightarrow$ Defines what the data is
  - Note: Presentation is handled separately
- Benefits
  - Define data once, transform for multiple outputs
  - Select and share only necessary data

Benefits of a Well-Used Data Dictionary

- Saves Time: Reduces ambiguity and rework, especially if started early
- “Single Source of Truth”: Central repository for all data definitions
- Resolves Disputes: Provides clear, agreed-upon meanings for data terms
- Aids Maintenance: Invaluable reference for understanding unfamiliar or legacy systems
- Foundation for Automation: Enables automated documentation, code generation, and validation checks